

Project Information			
Group ID	16		Semester (e.g., SS26)
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Project Title	Chat Application		

Project Description	This project focuses on developing a distributed chat application in Python. Multiple clients can join a shared chat room and exchange messages through replicated servers. The system is designed to be scalable and fault tolerant. The application supports dynamic discovery of hosts, leader election, ordered reliable multicast communication, and crash fault tolerance. Replicated servers maintain the same state to ensure system availability even if the leader server crashes. The system uses vector clocks and NACK-based reliable multicast to guarantee message ordering and delivery consistency between all participants.
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Architectural Model	Client-Server Architecture Model with replicated servers and leader-based coordination. - Clients communicate with the leader server. - The leader multicasts messages to all participants. - Replicated servers maintain synchronized state. - Heartbeats are used for crash detection and recovery.
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Dynamic Discovery	Dynamic discovery is implemented using multicast/broadcast communication. - New clients and servers can join without restarting the system. - Each participant maintains a local group view. - Broadcast sender/listener logic is used for discovering active participants dynamically.
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Fault Tolerance	The system handles crash failures of clients and servers. - Heartbeats are exchanged between replicated servers. - Missing heartbeats indicate server crashes. - Replica servers maintain replicated chat state. - Automatic failover ensures uninterrupted communication when the leader crashes.
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Voting	Voting -Leader election is implemented using the LCR (Le Lann – Chang – Roberts) algorithm. -One server acts as the leader. -The leader manages membership and message ordering. -If the leader crashes, a new leader is automatically elected among replicated servers.
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